

Introductory Calculus



Direct substitution

- 1 Find $\lim_{x \rightarrow 2} (3x^2 - 2x + 1)$.
 - 2 Find $\lim_{x \rightarrow -1} (x^3 - 5x)$.
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Indeterminate forms: factor and cancel

- 3 Find $\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5}$.
 - 4 Find $\lim_{x \rightarrow -3} \frac{x^2 - 9}{x + 3}$.
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The derivative as a limit

- 5 Write the difference-quotient definition of $f'(x)$.
 - 6 Using Theorem 5.1 ($f(x) = x^2 \Rightarrow f'(x) = 2x$), find the slope of the tangent line to $f(x) = x^2$ at $x = 3$.
 - 7 Using Theorem 5.1, find the slope of the tangent line to $f(x) = x^2$ at $x = -5$.
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Differential equations

- 8 Show that $y = x^2 - 7$ solves $\frac{dy}{dx} = 2x$.
 - 9 Explain why solving a differential equation gives a family of functions rather than a single number.
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Multiple choice — easy

10 Find $\lim_{x \rightarrow 0} (-2x^2 - 6x - 6)$.

- a. -6
- b. 6
- c. -10
- d. -2

11 Find $\lim_{x \rightarrow -4} (-3x^2 + 8x + 3)$.

- a. -81
- b. -73
- c. -77
- d. 77

12 Find $\lim_{x \rightarrow -5} (-3x^2 - 6x + 2)$.

- a. -43
- b. -39
- c. -47
- d. 43

13 Find $\lim_{x \rightarrow -1} (-x^2 - x - 5)$.

- a. -1
- b. 5
- c. -5
- d. -9

14 Find $\lim_{x \rightarrow -3} (x^2 - 4x + 5)$.

a. 22

b. 26

c. -26

d. 30

15 Find $\lim_{x \rightarrow 5} (-x^2 + 9x - 6)$.

a. -14

b. 18

c. 10

d. 14

16 Find $\lim_{x \rightarrow -3} (2x^2 + 5x - 1)$.

a. -2

b. 6

c. 8

d. 2

17 Find $\lim_{x \rightarrow 0} (3x^2 + 7x + 4)$.

a. 8

b. -4

c. 0

d. 4

18 Find $\lim_{x \rightarrow 4} (2x^2 + 5x + 9)$.

- a. 57
- b. -61
- c. 65
- d. 61

19 Find $\lim_{x \rightarrow -2} (-2x^2 + 6x + 9)$.

- a. -7
- b. -11
- c. 11
- d. -15

20 Find $\lim_{x \rightarrow 5} (3x^2 + x + 8)$.

- a. -88
- b. 92
- c. 88
- d. 84

21 Find $\lim_{x \rightarrow -5} (-2x^2 - 8x + 4)$.

- a. 6
- b. -10
- c. -6
- d. -2

22 Find $\lim_{x \rightarrow 1} (x^2 + 4x - 3)$.

- a. -2
- b. 2
- c. 8
- d. 6

23 Find $\lim_{x \rightarrow -2} (-2x^2 + 5x - 9)$.

- a. 27
- b. -23
- c. -27
- d. -31

24 Find $\lim_{x \rightarrow -5} (x^2 - 4x + 9)$.

- a. 50
- b. 54
- c. -54
- d. 58

Multiple choice — medium

25 Find $\lim_{x \rightarrow -7} \frac{x^2 + 15x + 56}{x + 7}$.

- a. 1
- b. -2
- c. -1
- d. 4

26 Find $\lim_{x \rightarrow -3} \frac{x^2 + 2x - 3}{x + 3}$.

a. -4

b. 4

c. -1

d. -7

27 Find $\lim_{x \rightarrow -5} \frac{x^2 + 6x + 5}{x + 5}$.

a. -1

b. -7

c. -4

d. 4

28 Find $\lim_{x \rightarrow -5} \frac{x^2 + 7x + 10}{x + 5}$.

a. 0

b. 3

c. -3

d. -6

29 Find $\lim_{x \rightarrow 5} \frac{x^2 - 11x + 30}{x - 5}$.

a. 1

b. -1

c. 2

d. -4

30 Find $\lim_{x \rightarrow -3} \frac{x^2 + 10x + 21}{x + 3}$.

a. 1

b. 7

c. -4

d. 4

31 Find $\lim_{x \rightarrow 2} \frac{x^2 - 9x + 14}{x - 2}$.

a. -5

b. -2

c. -8

d. 5

32 Find $\lim_{x \rightarrow 6} \frac{x^2 - 14x + 48}{x - 6}$.

a. -5

b. 2

c. -2

d. 1

33 Find $\lim_{x \rightarrow -5} \frac{x^2 - 25}{x + 5}$.

a. 10

b. -13

c. -10

d. -7

34 Find $\lim_{x \rightarrow 7} \frac{x^2 - 4x - 21}{x - 7}$.

- a. 13
- b. -10
- c. 7
- d. 10

35 Find $\lim_{x \rightarrow -2} \frac{x^2 - 2x - 8}{x + 2}$.

- a. -6
- b. -3
- c. 6
- d. -9

36 Find $\lim_{x \rightarrow -3} \frac{x^2 + x - 6}{x + 3}$.

- a. -2
- b. -8
- c. -5
- d. 5

37 Find $\lim_{x \rightarrow -3} \frac{x^2 + 9x + 18}{x + 3}$.

- a. 6
- b. 3
- c. -3
- d. 0

38 Find $\lim_{x \rightarrow 1} \frac{x^2 + 6x - 7}{x - 1}$.

- a. 5
- b. -8
- c. 8
- d. 11

39 Find $\lim_{x \rightarrow 0} \frac{x^2 + 8x}{x}$.

- a. 11
- b. 8
- c. 5
- d. -8

Multiple choice — hard

40 Given $f(x) = -x^2 + 5x - 4$, use the power rule ($\frac{d}{dx}x^2 = 2x$, $\frac{d}{dx}x = 1$) to find $f'(2)$.

- a. 1
- b. -1
- c. 3
- d. 4

41 Given $f(x) = 4x^2 + 5x + 9$, use the power rule ($\frac{d}{dx}x^2 = 2x$, $\frac{d}{dx}x = 1$) to find $f'(-6)$.

- a. -45
- b. 43
- c. -41
- d. -43

42 Given $f(x) = -3x^2 + 9x - 2$, use the power rule ($\frac{d}{dx}x^2 = 2x$, $\frac{d}{dx}x = 1$) to find $f'(3)$.

- a. -7
- b. -11
- c. 9
- d. -9

43 Given $f(x) = 4x^2 + 3x + 7$, use the power rule ($\frac{d}{dx}x^2 = 2x$, $\frac{d}{dx}x = 1$) to find $f'(4)$.

- a. 35
- b. -35
- c. 33
- d. 37

44 Given $f(x) = -x^2 - 3x - 1$, use the power rule ($\frac{d}{dx}x^2 = 2x$, $\frac{d}{dx}x = 1$) to find $f'(0)$.

- a. -1
- b. -5
- c. -3
- d. 3

45 Find $\lim_{x \rightarrow \infty} \frac{6x + 3}{x - 9}$.

- a. $\frac{13}{2}$
- b. -6
- c. $\frac{11}{2}$
- d. 6

46 Find $\lim_{x \rightarrow \infty} \frac{3x + 1}{-5x - 6}$.

a. $-\frac{1}{10}$

b. $\frac{3}{5}$

c. $-\frac{11}{10}$

d. $-\frac{3}{5}$

47 Find $\lim_{x \rightarrow \infty} \frac{x + 5}{3x - 8}$.

a. $\frac{5}{6}$

b. $\frac{1}{3}$

c. $-\frac{1}{3}$

d. $-\frac{1}{6}$

48 Find $\lim_{x \rightarrow \infty} \frac{-2x}{-5x + 2}$.

a. $-\frac{2}{5}$

b. $\frac{2}{5}$

c. $\frac{9}{10}$

d. $-\frac{1}{10}$

49 Find $\lim_{x \rightarrow \infty} \frac{-4x - 2}{-4x + 4}$.

a. $\frac{3}{2}$

b. 1

c. $\frac{1}{2}$

d. -1

- 50 Find the average rate of change of $f(x) = x^2$ on the interval from $x = -1$ to $x = 0$.
- a. -3
 - b. 3
 - c. 1
 - d. -1
- 51 Find the average rate of change of $f(x) = x^2$ on the interval from $x = 3$ to $x = 1$.
- a. 2
 - b. 6
 - c. -4
 - d. 4
- 52 Find the average rate of change of $f(x) = x^2$ on the interval from $x = -1$ to $x = 1$.
- a. 4
 - b. -2
 - c. 2
 - d. 0
- 53 Find the average rate of change of $f(x) = x^2$ on the interval from $x = 3$ to $x = 4$.
- a. 5
 - b. -7
 - c. 9
 - d. 7

54 Find the average rate of change of $f(x) = x^2$ on the interval from $x = 2$ to $x = -1$.

a. 3

b. 5

c. 1

d. -1